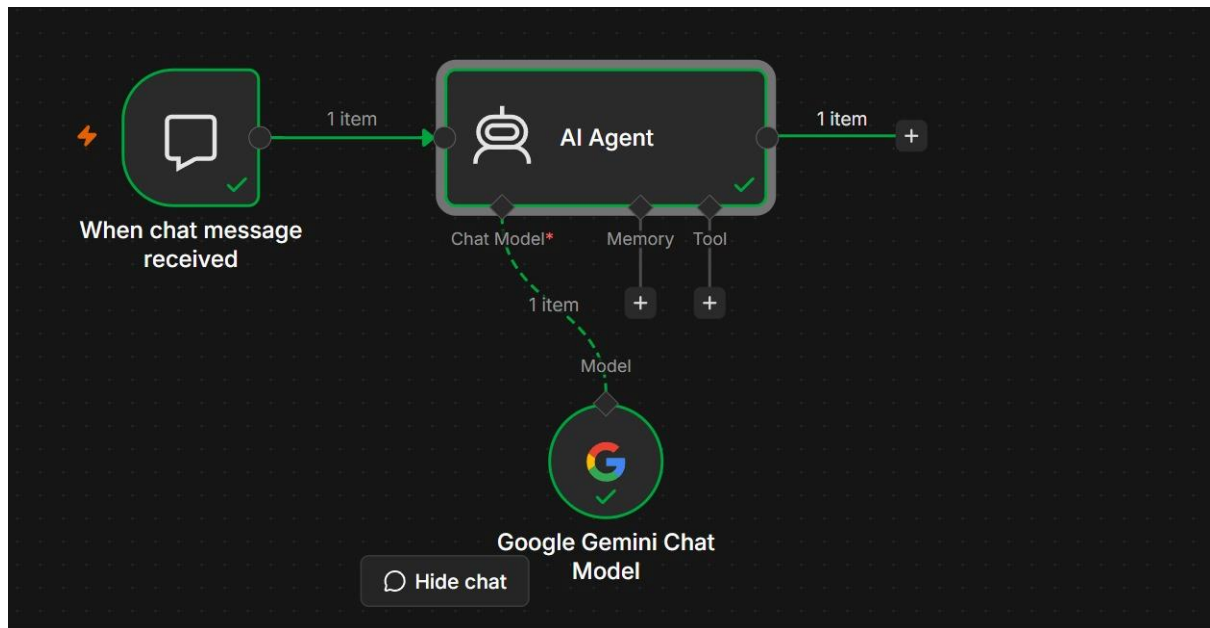


PROJECT-26

AI Cover Letter Generator



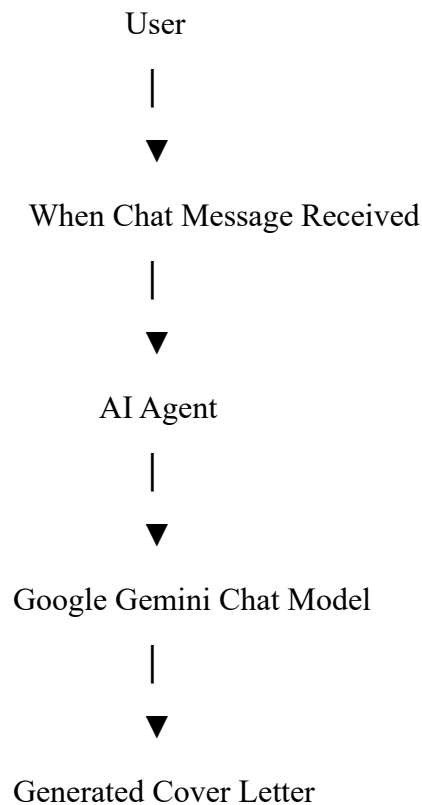
Project Objective

The objective of this project is to build an **AI-powered Cover Letter Generator** that creates professional, personalized, and job-specific cover letters based on the user's job role, experience, skills, and company details. The AI Agent generates compelling cover letters that highlight the candidate's qualifications and increase the chances of securing interviews.

The AI Agent can generate:

- Software Engineer cover letters
- Data Analyst cover letters
- Marketing cover letters
- HR cover letters
- Internship cover letters
- Fresher cover letters
- Experienced professional cover letters
- Career change cover letters
- Remote job cover letters
- Customized cover letters for any industry

Workflow Architecture



Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's job application details
2	AI Agent	Understands the user's profile and prepares the cover letter prompt
3	Google Gemini Chat Model	Generates a personalized cover letter

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
- Google AI Studio API Key
- Google Gemini Chat Model configured
- Internet connection

Step 1 – Create Workflow

Open **n8n**

Click

New Workflow

Rename it

AI Cover Letter Generator

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's job application details.

Example Input

Write a cover letter for a Software Engineer position at Google. I have 3 years of experience in Python, Java, Machine Learning, and Cloud Computing.

Output

```
{  
  "chatInput": "Write a cover letter for a Software Engineer position at Google. I have 3 years  
of experience in Python, Java, Machine Learning, and Cloud Computing."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the **System Message**.

You are an AI Cover Letter Generator.

Your task is to create professional, personalized, and persuasive cover letters based on the user's information.

Instructions:

- Understand the user's target job role, company, skills, qualifications, and experience.
 - Write a professional and engaging cover letter.
 - Begin with a formal greeting.
 - Introduce the applicant and mention the job position.
 - Highlight relevant skills, experience, and achievements.
 - Explain why the applicant is interested in the company.
 - Conclude with a confident closing statement and thank the employer.
 - Maintain a professional and positive tone.
 - Keep the cover letter between 250 and 400 words unless the user requests otherwise.
 - Return only the completed cover letter.
-

Step 5 – Add Google Gemini Chat Model

Click

+ **Chat Model**

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini
Model	Gemini 2.5 Flash (or another available Gemini model)
Authentication	Google AI Studio API Key

Connect it to the **AI Agent**.

Final Workflow

When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Write a cover letter for a Data Analyst position at Microsoft. I have two years of experience in SQL, Python, Power BI, and Excel.

↓

Step 2

The **When Chat Message Received** node receives the user's request.

↓

Step 3

The **AI Agent** analyzes the job role, company, skills, and experience, then prepares the prompt.

↓

Step 4

The prompt is sent to the **Google Gemini Chat Model**.

↓

Step 5

Gemini generates a professional and personalized cover letter.

↓

Step 6

The generated cover letter is returned to the user.

Sample Inputs and Outputs

Example 1

Input

Write a cover letter for a Software Engineer position.

Output

Dear Hiring Manager,

I am excited to apply for the Software Engineer position at your organization. With three years of experience in software development and strong skills in Python, Java, and cloud technologies, I am confident in my ability to contribute effectively to your development team.

Throughout my career, I have built scalable applications, collaborated with cross-functional teams, and delivered high-quality software solutions. I am passionate about solving complex problems and continuously learning new technologies.

I would welcome the opportunity to discuss how my skills and experience align with your organization's goals. Thank you for your time and consideration.

Sincerely,

John Doe

Example 2

Input

Generate a cover letter for a Data Analyst role.

Output

Dear Hiring Manager,

I am writing to express my interest in the Data Analyst position. My experience with SQL, Python, Excel, and Power BI has enabled me to analyze complex datasets and provide actionable business insights.

I enjoy transforming data into meaningful reports that support informed decision-making. I am eager to contribute my analytical skills to your organization and continue growing professionally.

Thank you for considering my application.

Sincerely,

John Doe

Example 3

Input

Write a cover letter for a Marketing Executive position.

Output

Dear Hiring Manager,

I am pleased to apply for the Marketing Executive position. With experience in digital marketing, social media management, SEO, and content strategy, I have successfully planned and executed campaigns that increased customer engagement and brand visibility.

I am excited about the opportunity to bring creativity, analytical thinking, and strategic planning to your marketing team.

Thank you for your valuable time.

Sincerely,

John Doe

Example 4

Input

Generate a cover letter for a fresher applying for an internship.

Output

Dear Hiring Manager,

I am excited to apply for the internship opportunity at your organization. As a recent graduate with strong academic knowledge and hands-on project experience, I am eager to begin my professional career.

I am a quick learner with excellent communication and teamwork skills. I believe this internship will allow me to apply my knowledge while learning from experienced professionals.

Thank you for considering my application. I look forward to the opportunity to contribute to your organization.

Sincerely,

John Doe

Workflow Diagram

User

|



When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

|



Generated Cover Letter

Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive the user's request.

Solution

Verify that the **Prompt** field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
 - Check your API quota.
 - Retry the request.
-

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered Cover Letter Generator in n8n.
 - Configure an AI Agent with Google Gemini.
 - Design prompts for professional document generation.
 - Generate personalized cover letters automatically.
 - Build conversational AI workflows for career assistance.
-

Real-World Applications

- Job application assistance
- Career coaching
- Internship applications
- Resume and cover letter preparation
- HR support tools

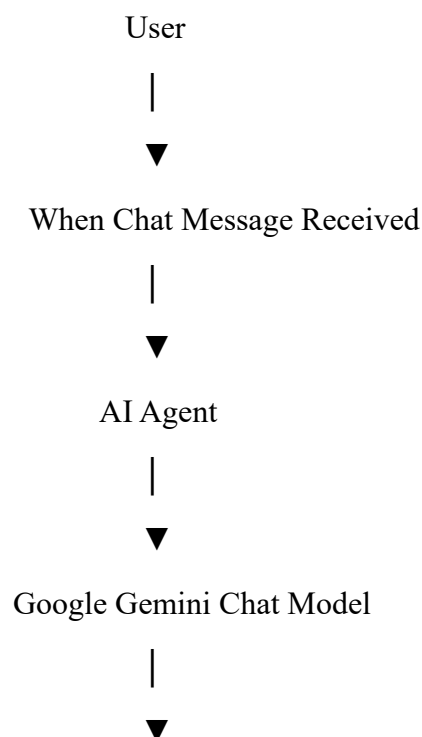
- University placement services
 - Recruitment assistance
 - Professional profile enhancement
 - Employment portals
 - AI-powered career guidance
-

Practice Challenge

Enhance this project by adding:

1. A job role selector (Software Engineer, Data Analyst, HR Manager, Marketing Executive, etc.).
 2. A company name input field for personalized cover letters.
 3. A tone selector (Professional, Formal, Friendly, Confident).
 4. A Memory node to remember the user's preferred writing style and job role.
 5. A Google Sheets node to save generated cover letters.
 6. A PDF Generator or Email node to automatically generate and send the cover letter in PDF format.
-

Final Workflow



Expected Result

The **AI Cover Letter Generator** accepts the user's job role, company name, skills, qualifications, and experience, uses the **Google Gemini Chat Model** to generate a personalized, professional, and persuasive cover letter, and returns a well-structured document ready for job applications. This project teaches prompt engineering, AI-powered document generation, and conversational workflow design in n8n, making it a strong foundation for building intelligent career assistance and recruitment automation solutions.